

Wealthy British aristocrat Henry Cavendish was an outstanding theoretical chemist and physicist whose meticulous research established the true nature of air and water and the properties of hydrogen and carbon dioxide. He also made the first calculation of the density, and hence the weight, of the Earth.

MILESTONES

BEGINS EXPERIMENTS

After leaving college in 1751, lives with his father, Lord Cavendish, and sets up his own home laboratory.

WINS PRIZE FOR PAPER

Receives Royal Society's Copley Medal in 1766 for his paper on properties of gases, including hydrogen.

NATURE OF WATER

Announces to Royal Society that water is a compound of hydrogen and oxygen in 1784.

INVESTIGATES AIR

In 1785, runs experiments to analyze the composition of air and finds another gas present.

"WEIGHS" THE WORLD

Becomes the first person to measure the force of gravity between masses; reports his results in 1798.

Henry Cavendish studied natural philosophy (science) at Cambridge University and then settled in London, where he played an active role in scientific organizations, particularly the Royal Society, which he joined in 1760. He started conducting experiments in 1764, exploring many areas of science—including electricity, chemistry, and physics—although he published only a few of his findings during his 50-year career.

Isolating hydrogen

In the 18th century, chemists started to investigate ways of generating and separating various gases, although they thought these substances were different types of air rather than unique chemicals. Cavendish's first publication, in 1766, was a combination of three chemistry papers on "factitious airs," or gases, created in a laboratory. In one of the rigorous investigations that characterized his work, he had dissolved zinc, iron, and tin in acids and found this produced a gas that he called "inflammable air." It was ultimately given the chemical name hydrogen. The most abundant element

In 1783, after studying the gas chemistry research of Cavendish and others, Parisian Jacques Charles developed the world's first hydrogen-filled balloon, starting a ballooning craze.

